Imaging and Management of Radial Scars and Complex Sclerosing Lesions

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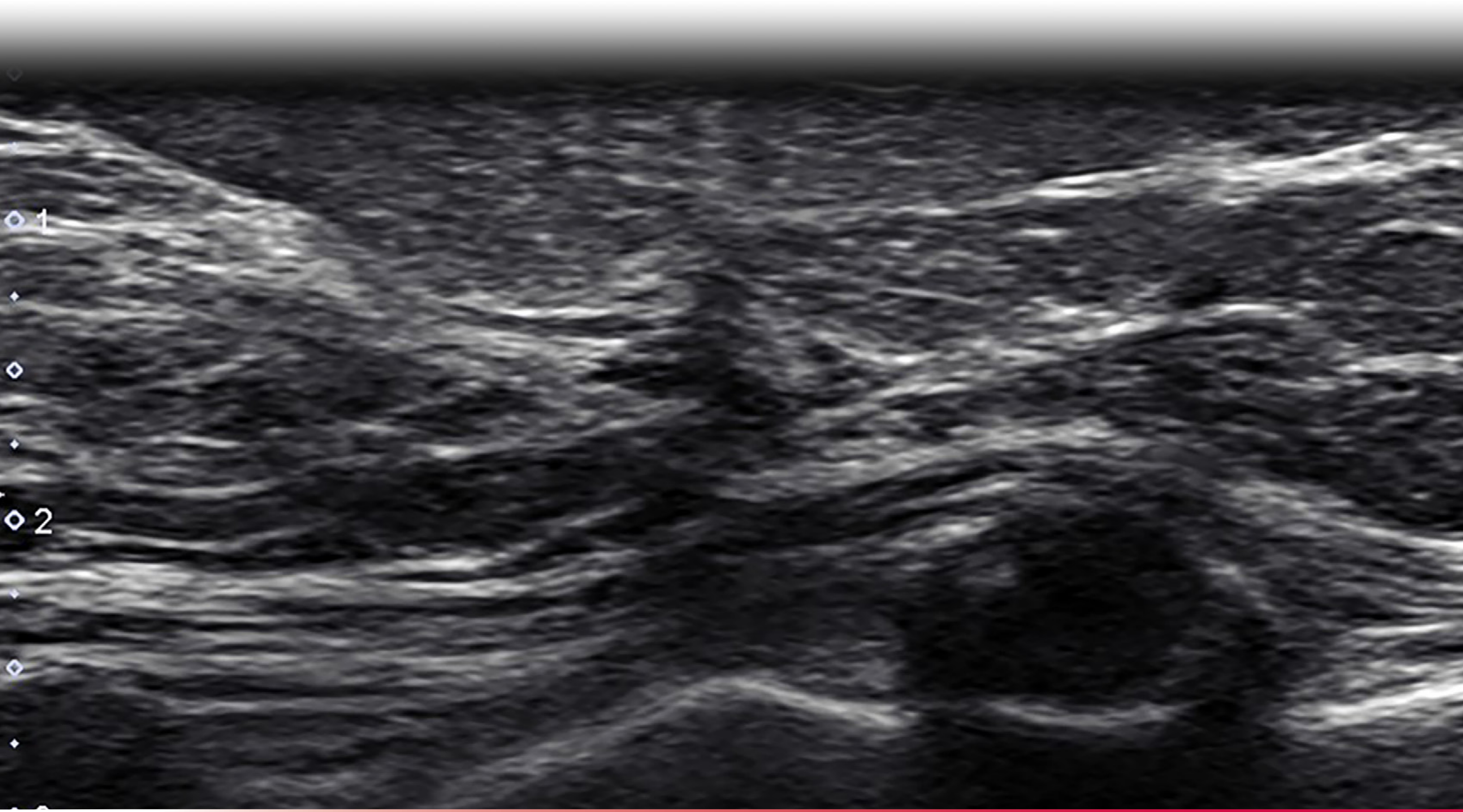
Radial scars and complex sclerosing lesions, often collectively referred to as radial sclerosing lesions (RSLs), are breast lesions characterized by sclerotic stroma with entrapped epithelial elements. RSLs have imaging features that overlap with those of breast malignancy and often become the target of imaging-guided biopsy given their suspicious imaging appearance. These can be identified in isolation or can also be associated with atypia or other high-risk lesions that have intrinsic malignant potential, increasing the risk of carcinoma and affecting prognosis and management of RSLs. Because of this, management of these lesions remains controversial. Traditional management has been surgical excisional biopsy. However, as more RSLs are identified (because digital breast tomosynthesis allows identification of more architectural distortions), optimal management is evolving. Physicians in some practices are using a multidisciplinary approach to the management of RSLs when deciding on surgical excision of these lesions versus imaging follow-up. These discussions also incorporate individual patient risk factors and greater patient informed medical decision making. Reported upgrade rates of RSLs at core needle biopsy vary and can depend on the sampling method, number of samples, gauge of the needle, target being sampled, and radiologic–pathologic concordance or discordance. A precise sampling technique also allows greater accuracy of diagnosis and lower upgrade rates for these lesions, with radiologic–pathologic correlation as an integral component for further management decisions. The authors review the overall histopathologic, clinical, and imaging features of RSLs and discuss appropriate management based on currently available data regarding upgrade rates.



CONSENSUS ON RADIOLOGY-PATHOLOGY CORRELATION

Quiz questions for this article are available through the Online Learning Center.

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Abbreviations: CNB = core needle biopsy, CSL = complex sclerosing lesion, DBT = digital breast tomosynthesis, DCIS = ductal carcinoma in situ, RSL = radial sclerosing lesion, TC = tubular carcinoma

TEACHING POINTS

- RSLs are often found incidentally at mammographic screening.
- The imaging features of RSLs often overlap with those of breast malignancy; as a result, sampling is required to exclude malignancy.
- The mammographic features of RSLs are characterized by a stellate or spiculated appearance with architectural distortion and central lucency.
- Multiple studies of RSLs without atypia have shown upgrade rates ranging from 0% to 2.2%.
- Breast radiologists play a critical role in patient treatment recommendations and must be mindful that surgical excision for all RSLs, particularly those without associated atypia, may result in overtreatment of patients. Thus, imaging follow-up may be a reasonable approach for patients with accurately sampled RSLs without atypia.

Introduction

Radial sclerosing lesions (RSLs) are noncancerous breast lesions characterized by fibroelastosis with entrapment of epithelial elements in the stroma. RSLs include both radial scars and complex sclerosing lesions (CSLs) that are differentiated by their histopathologic architecture and, by some authors, by their size. Radial scars have a central fibroelastic core with ducts radiating from the center, resulting in a stellate appearance of the lesion at imaging and pathologic examination. CSLs are larger and have a more complex and disorganized architecture. Often, however, the terms radial scar, CSL, and RSL are used interchangeably.

RSLs are often found incidentally at mammographic screening. The imaging features of RSLs often overlap with those of breast malignancy; as a result, sampling is required to exclude malignancy. At the time of pathologic review, they may be associated with other high-risk lesions, such as lobular neoplasia, or atypia, which increases the risk of carcinoma, if present. Thus, these lesions are often excised surgically to exclude an associated malignancy. The number of diagnosed RSLs has increased with the introduction of digital breast tomosynthesis (DBT) into routine clinical practice. Authors of one study (1) found that 33% of DBT–detected architectural distortions are radial scars; in comparison, 12% of digital mammography–detected architectural distortions are radial scars. Thus, the evidence related to RSLs and accepted treatment strategies must be reviewed.

The purposes of this article are to review the overall histopathologic, clinical, and imaging features of RSLs and to discuss appropriate management.

Pathologic Evaluation

Definitions and Terminology

Breast lesions composed of benign epithelial elements entrapped and distorted by fibroelastosis and hyalinized stroma are pathologically characterized as sclerosing lesions. The *World Health Organization Classification of Tumors* (2) notes that in addition to radial scar, CSL, and RSL, the term *benign sclerosing ductal proliferation* is also acceptable. Pathologically, the distinction between a radial scar and a CSL is based on the overall architecture of the lesion. In the literature, these may be grouped together and referred to as RSLs. RSLs may be incidental findings in breast tissue removed for other reasons, or they may be detected radiologically and may be the target of a core needle biopsy (CNB).

Radial scars are small and characterized by a sclerotic central nidus with epithelial elements radiating like spokes from the center, which imparts a stellate appearance (Fig 1A). CSLs are larger and have a disorganized appearance (Fig 1B). In both, associated proliferative changes such as usual ductal hyperplasia, papillomas, apocrine metaplasia, adenosis, and cysts may or may not be present. Calcifications may also be associated with these lesions (3,4). The overall architecture is important to distinguish between radial scars and CSLs; in small biopsy samples, this may not be possible, and a descriptive diagnosis or a diagnosis of a “sclerosing lesion” or an RSL may be rendered.

In radial scars and CSLs, the glandular elements are distorted by the elastotic and sclerotic stroma, which can impart an infiltrative appearance. A myoepithelial cell layer may not be evident on routine hematoxylin and eosin–stained sections, raising concern for an invasive carcinoma. In histologically problematic cases, using stains for myoepithelial cells such as p63 and smooth muscle myosin heavy chain can highlight the myoepithelial layer around the benign epithelial elements. The myoepithelial cells of sclerosing lesions may show a focal reduction or absence of expression of these markers (5).

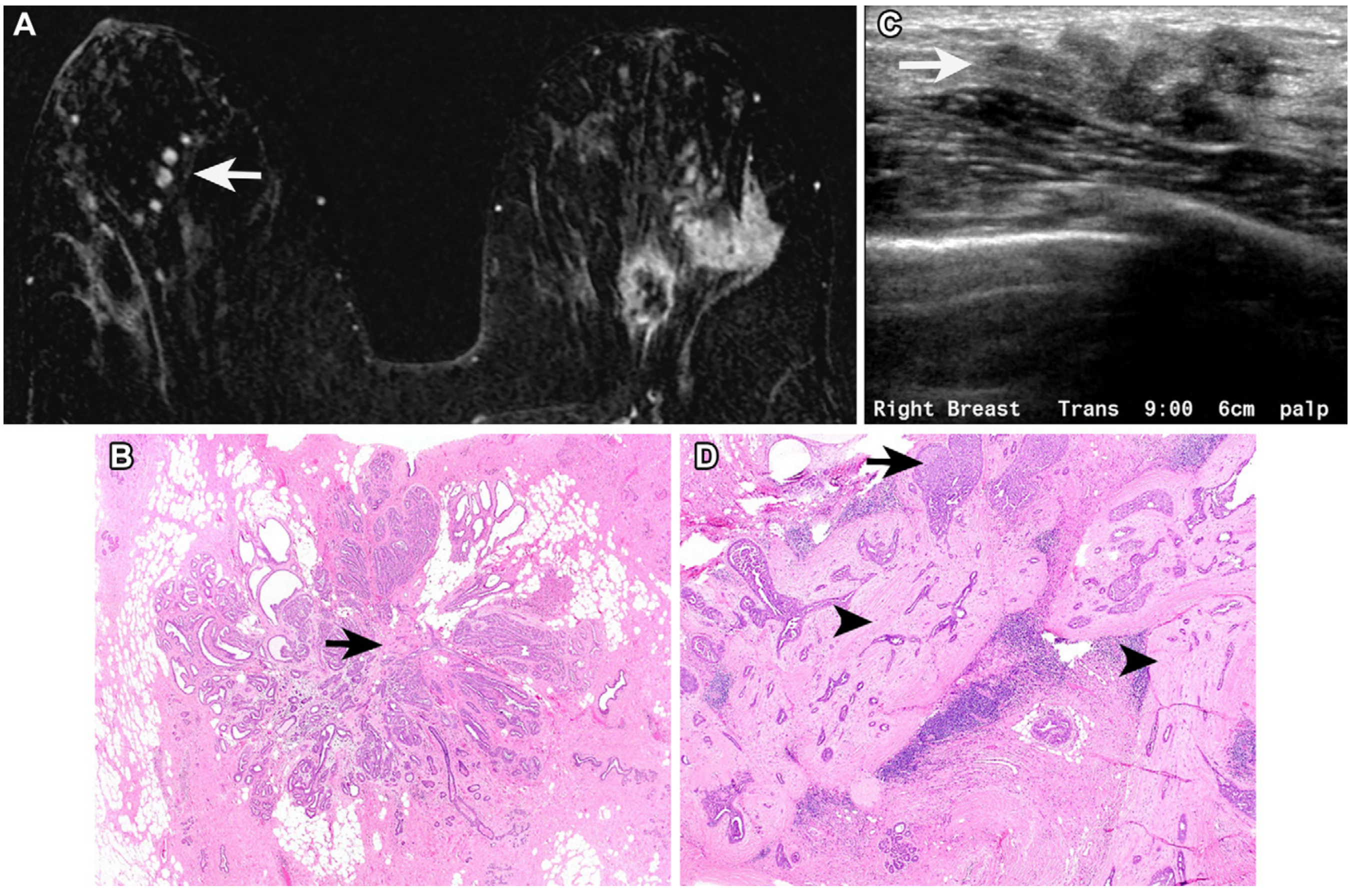
Differential Diagnosis

Tubular carcinomas (TCs) have morphologic similarities to RSLs in that they are both small glandular proliferations; however, TCs are an invasive carcinoma and RSLs are benign. Radial scars are the main differential diagnostic consideration of a TC (Fig 2). TCs are composed of small round to oval glands and tubules embedded in a dense fibroelastotic stroma, often with a stellate architecture, similar to radial scars (2). The glands in TCs lack myoepithelial cells, whereas they are present around the nests in RSLs. The epithelial cells have an apical snout, which distinguishes them from low-grade carcinomas of no special type (also known as invasive ductal carcinomas). A majority of TCs have associated lesions in the low-grade neoplasia pathway, which include columnar cell lesions, flat epithelial atypia, atypical ductal hyperplasia, low-grade ductal carcinoma in situ, and lobular neoplasia (6).

Association with Malignancy and Atypia

RSLs are benign lesions, but associations with certain malignancies or atypia have been reported. Metaplastic carcinomas are rare, accounting for 0.2%–1% of all invasive breast carcinomas. Some cases of low-grade adenosquamous carcinomas or fibromatosis-like metaplastic carcinomas, which are uncommon variants of metaplastic carcinoma, have arisen

Figure 1. RSLs in two patients. **(A, B)** MR image **(A)** shows a radial scar in the right breast in a 33-year-old woman with a new diagnosis of left breast invasive carcinoma. MRI was performed to evaluate the extent of disease. Incidental enhancing foci are noted in a linear configuration in the central mid right breast (arrow in **A**). MRI-guided biopsy was performed and subsequently followed by needle-localized surgical excisional biopsy. Photomicrograph of a histologic specimen **(B)** shows a lesion with stellate architecture and a sclerotic central nidus (arrow in **B**). Small distorted benign glands and tubules are entrapped in the sclerotic stroma. Ducts radiate from the center of the lesion. (Hematoxylin-eosin stain; original magnification ×40). **(C, D)** Transverse US image **(C)** in a 28-year-old woman with a 3-month history of a palpable mass in the lower outer quadrant of the right breast shows a 2.1-cm, irregular, heterogeneous, hypoechoic CSL (arrow in **C**). This lesion demonstrates a more complex and disorganized architecture than that of an RS. Histologic specimen **(D)** shows that glands are embedded in and distorted by the dense pink sclerotic stroma (arrowheads). Some of the glands are expanded by usual ductal hyperplasia (arrow in **D**). (Hematoxylin-eosin stain; original magnification ×40.)



in association with RSLs and sclerosing papillomas (2,7–9). Low-grade adenosquamous carcinomas are characterized by infiltrating small glands and tubules admixed with nests of squamous cells (2). Fibromatosis-like metaplastic carcinomas are composed of infiltrative bland spindle cells that resemble desmoid fibromatosis (1). There have also been rare reports of TCs in association with RSLs (10,11). RSLs associated with or involving atypical hyperplasia and in situ or invasive carcinomas are seen more often in lesions that are larger than 0.6 cm, are mammographically detected, and tend to occur in women older than 50 years (2).

Pathogenesis

The pathogenesis of RSL is uncertain. They are not associated with previous surgery or trauma; thus, the term *radial scar* is a misnomer. Because of the histologic overlap between sclerosing papillomas and CSLs, some CSLs are thought to represent a later stage of sclerosing papillomas (9). The pathogenesis of RSLs has not been shown to be associated with menopausal status, parity, contraceptive use, or treatment with tamoxifen (12). Although postmenopausal status is not associated with the initial pathogenesis of RSLs, demographic features including postmenopausal status and age older than 50 years are associated with upgrading of RSL at surgical excision (13,14).

Clinical Presentation

RSLs are most commonly found in asymptomatic women aged 30–60 years (15). An RSL is not usually palpable or associated with skin thickening or skin retraction. The frequency of RSLs in autopsy series (12,16) of unselected breasts in female patients ranged from 14% to 28%, emphasizing that RSL is a common incidental histopathologic finding. However, an RSL is usually an incidental microscopic finding in surgical excision specimens or an incidental finding at mammography. In a retrospective study, Phantana-Angkool et al (17) found a statistically significant increase in the rate of RSLs diagnosed with use of CNB after the introduction of DBT: the rate increased from 0.04% to 0.13% ($P < .0001$). Although the rate of detection and diagnosis is expected to increase with the introduction and implementation of DBT screening programs, RSLs remain uncommon after image-guided biopsy.

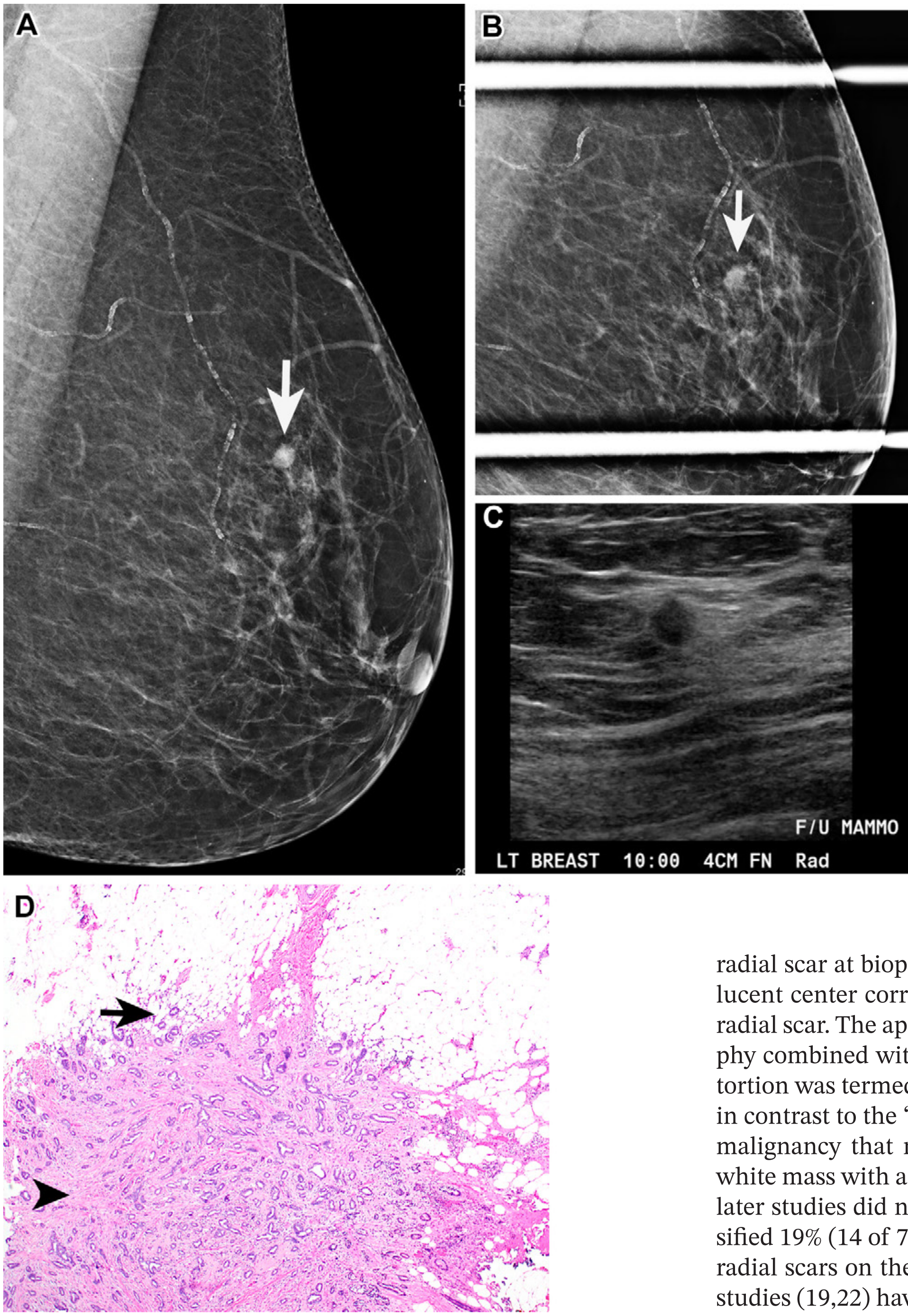


Figure 2. TC in a 67-year-old woman with a mass in the upper left breast. **(A–C)** Mediolateral oblique mammograms **(A, B)** and US image **(C)** show a 0.6-cm oval, microlobulated, hypoechoic, antiparallel mass with an echogenic rim (arrow in **A** and **B**). **(D)** Low-power photomicrograph of a histologic specimen shows the fibroelastotic desmoplastic stroma (arrowhead) with small round to oval glands and tubules embedded in it. The glands infiltrate into the surrounding adipose tissue (arrow). Myoepithelial cells are not present around the epithelial nests, which can be confirmed at immunohistochemical analysis. Focal atypical ductal hyperplasia was present in the background. (Hematoxylin-eosin stain; original magnification, ×40.)

Imaging Manifestations of RSLs

Mammography

RSLs may be multiple or bilateral in distribution. In one study (18) including 83 female cadavers at autopsy without known cancer found that 28% of women had a radial scar; 67% were multicentric and 43% were bilateral. Some authors differentiate radial scar from CSL at imaging by lesion size; radial scars are smaller than 1 cm and CSLs are larger than 1 cm (19). The mammographic features of RSLs are characterized by a stellate or spiculated appearance with architectural distortion and central lucency (Fig 3A). Studies (1) have shown that one-third of cases of distortion found at DBT corresponded to radial scar at biopsy. Historically, it was thought that a radiolucent center corresponding to central fat is characteristic of radial scar. The appearance of central black fat at mammography combined with the radiopaque white spicules of the distortion was termed the “black star” appearance (20). This was in contrast to the “white star” appearance attributed to breast malignancy that manifested as a central highly attenuating white mass with associated architectural distortion. However, later studies did not bear that out. Mitnick et al (21) misclassified 19% (14 of 73) of nonpalpable spiculated carcinomas as radial scars on the basis of a radiolucent center. More recent studies (19,22) have shown that a radiolucent center is not reliable in differentiating a radial scar from breast carcinoma (Fig 3B). This feature does not have a strong enough prospective performance to allow benign and malignant distortion to be distinguished (23). Thus, biopsy is required to differentiate RSLs and cancers.

Less commonly, RSL may appear at mammography as a mass or calcifications (24,25) (Fig 4). However, one study (26) of 53 patients actually demonstrated that a majority of RSLs were associated with masses, especially when deemed to be the target lesion rather than incidentally identified at the time of biopsy. In this study (26), all cases of RSL were classified as concordant by three study pathologists, and there were no associated high-risk lesions warranting excision. Architectural distortion was the second most commonly associated imaging finding, followed by calcifications. The only upgrade was

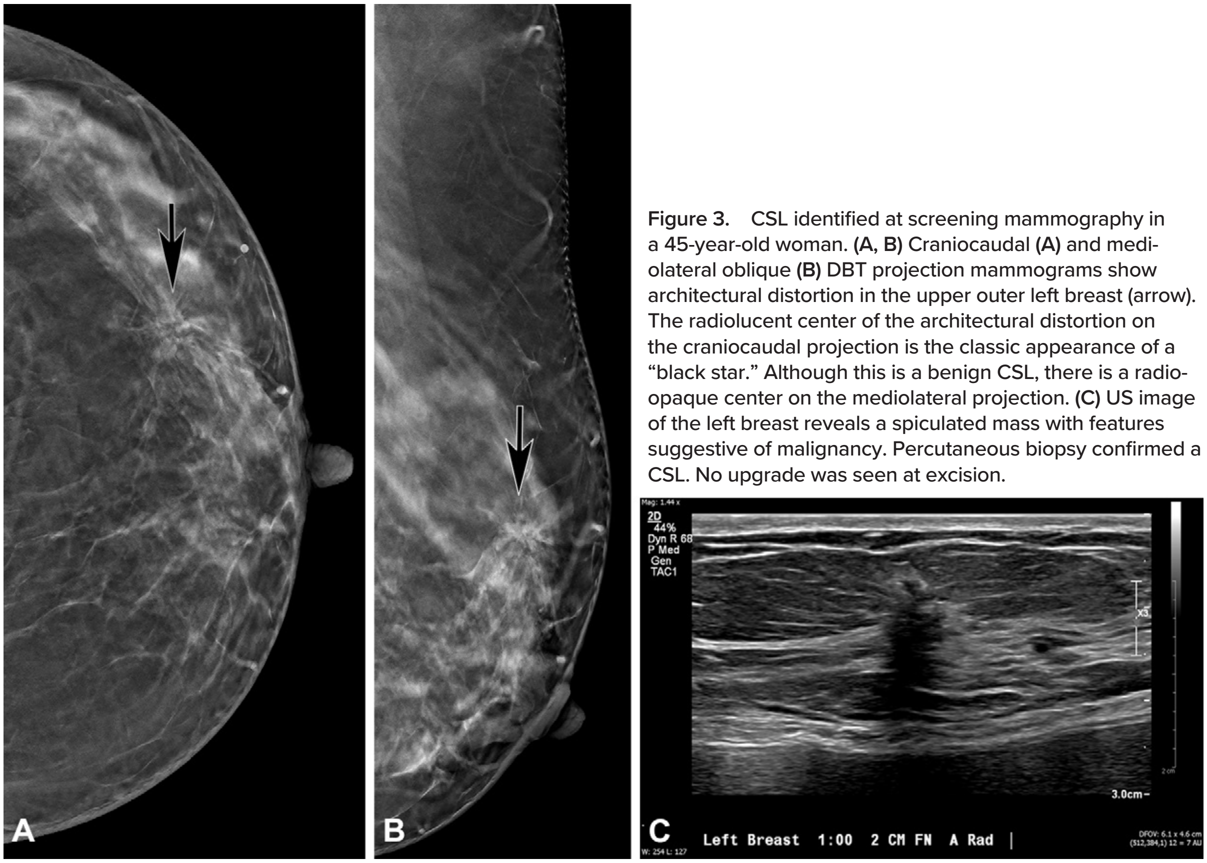
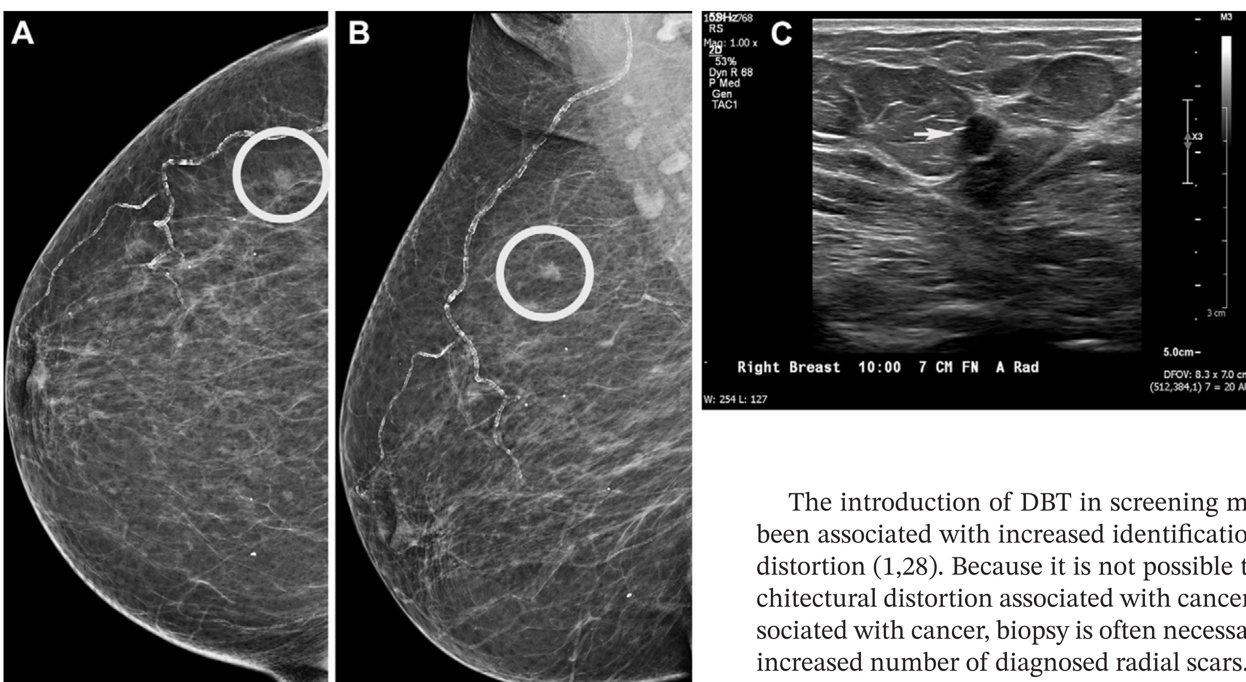


Figure 3. CSL identified at screening mammography in a 45-year-old woman. **(A, B)** Craniocaudal **(A)** and mediolateral oblique **(B)** DBT projection mammograms show architectural distortion in the upper outer left breast (arrow). The radiolucent center of the architectural distortion on the craniocaudal projection is the classic appearance of a "black star." Although this is a benign CSL, there is a radio-opaque center on the mediolateral projection. **(C)** US image of the left breast reveals a spiculated mass with features suggestive of malignancy. Percutaneous biopsy confirmed a CSL. No upgrade was seen at excision.

Figure 4. CSL in a 63-year-old woman. Craniocaudal **(A)** and mediolateral oblique **(B)** mammographic projections show a mass in the upper outer right breast (circle). US image of the right breast **(C)** shows an irregular hypoechoic mass (arrow). Percutaneous biopsy results confirmed a CSL. Atypical ductal hyperplasia was identified at excision.



associated with pleomorphic calcifications. The presence of calcifications may also suggest coexisting proliferative lesions, such as atypical ductal hyperplasia, lobular neoplasia, adenosis, or malignancy (27).

The introduction of DBT in screening mammography has been associated with increased identification of architectural distortion (1,28). Because it is not possible to differentiate architectural distortion associated with cancer from that not associated with cancer, biopsy is often necessary, resulting in an increased number of diagnosed radial scars. In addition, with the transition from two-dimensional stereotactic biopsy to DBT-guided biopsy, the percentage of biopsies performed for architectural distortion has increased from 2% to 17.7% (29). Of note, despite the increased detection of RSL with DBT, one study (17) showed no difference in the rate of upgrade to malignancy before and after the introduction of DBT.

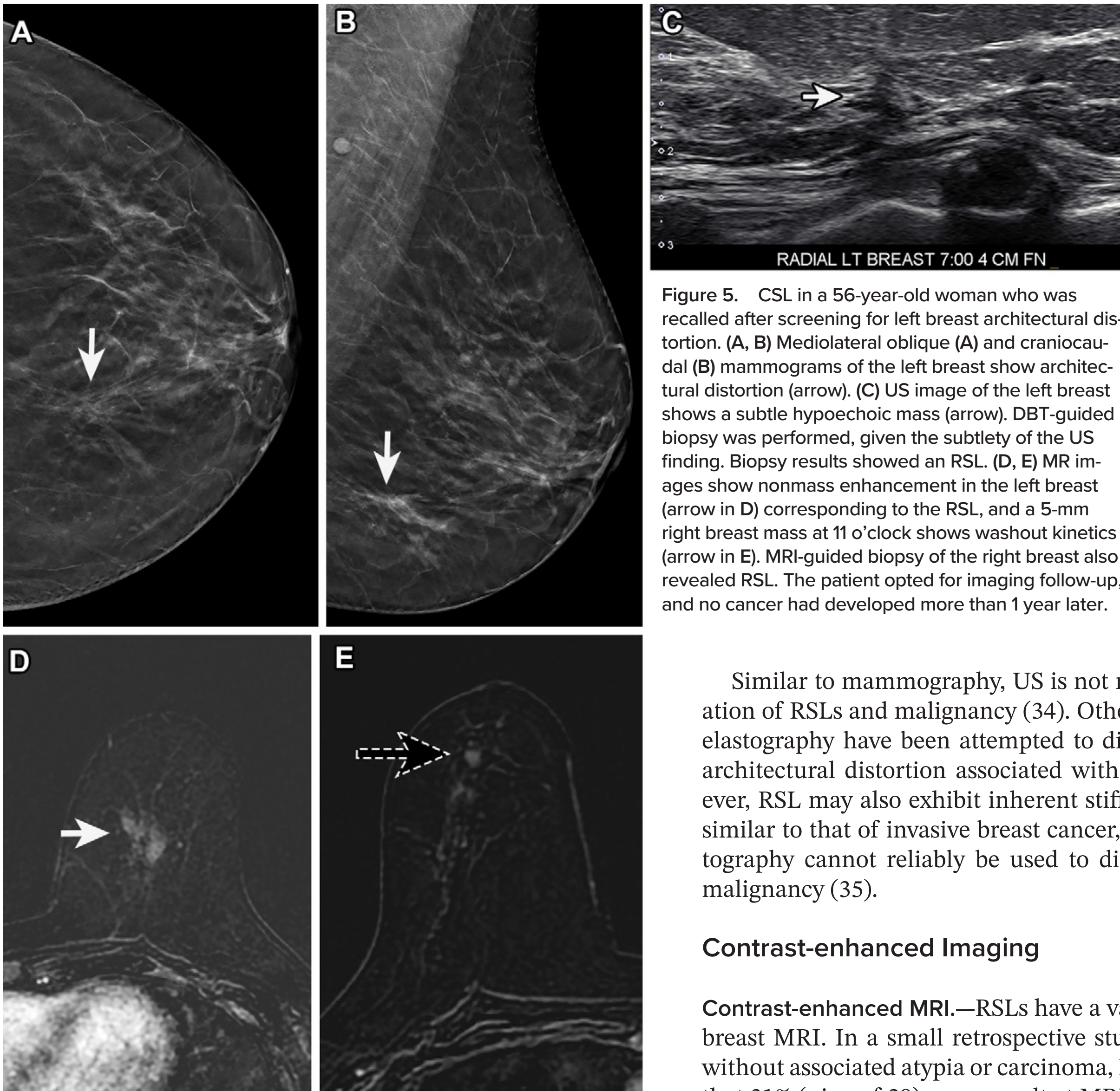


Figure 5. CSL in a 56-year-old woman who was recalled after screening for left breast architectural distortion. **(A, B)** Mediolateral oblique **(A)** and craniocaudal **(B)** mammograms of the left breast show architectural distortion (arrow). **(C)** US image of the left breast shows a subtle hypoechoic mass (arrow). DBT-guided biopsy was performed, given the subtlety of the US finding. Biopsy results showed an RSL. **(D, E)** MR images show nonmass enhancement in the left breast (arrow in **D**) corresponding to the RSL, and a 5-mm right breast mass at 11 o'clock shows washout kinetics (arrow in **E**). MRI-guided biopsy of the right breast also revealed RSL. The patient opted for imaging follow-up, and no cancer had developed more than 1 year later.

Ultrasonography

RSLs are frequently not visible at US. Abnormalities on US images are often found during targeted evaluation after detection of a mammographic abnormality, rather than as a primary US finding that is mammographically occult. In one study (30), 25 of 58 radial scars demonstrated a mass at US; the remaining 33 had no definite US abnormality. When RSLs are identified at US, they may appear as a hypoechoic irregular mass with indistinct margins, which may or may not exhibit posterior acoustic shadowing (Fig 3C). This appearance mimics malignancy, and biopsy is indicated. If an abnormality is seen at US, biopsies can be performed with US guidance. Bahl et al (1) demonstrated that if architectural distortion is seen at both mammography and US, the likelihood of malignancy is higher. Of note, because RSL is frequently not visible at US, the lack of a US correlate does not obviate the need for biopsy if an area of architectural distortion is seen mammographically. Studies (28,31–33) have shown that biopsy of architectural distortions seen only at DBT yielded cancer in 10%–47% of cases.

Similar to mammography, US is not reliable for differentiation of RSLs and malignancy (34). Other techniques such as elastography have been attempted to differentiate RSL from architectural distortion associated with breast cancer. However, RSL may also exhibit inherent stiffness at elastography, similar to that of invasive breast cancer, suggesting that elastography cannot reliably be used to differentiate RSL from malignancy (35).

Contrast-enhanced Imaging

Contrast-enhanced MRI.—RSLs have a variable appearance at breast MRI. In a small retrospective study ($n = 29$) of RSLs without associated atypia or carcinoma, Linda et al (36) found that 31% (nine of 29) were occult at MRI, whereas 69% (20 of 29) had an MRI correlate. The most common MRI appearance for RSLs in this study was an irregular enhancing mass, which in some cases showed associated architectural distortion (Fig 5). Other appearances at MRI included an enhancing focus, architectural distortion without an associated mass, and nonmass enhancement. When they are visible at MRI, RSLs show variable enhancement kinetics. Of the 20 excised RSLs with an MRI correlate, 50% demonstrated benign persistent enhancement kinetics, 11% demonstrated plateau enhancement kinetics, and 39% demonstrated suspicious washout enhancement kinetics. Of the nine (45%) lesions demonstrating benign enhancement kinetics, eight were assigned a Breast Imaging Reporting and Data System category of 3, 4, or 5. Linda et al (36) also showed that the imaging correlates at mammography and US of RSLs exhibited morphologic features similar to those seen at MRI.

Some centers use MRI for triage of patients with biopsy-proven RSLs to excision or surveillance. Evidence suggests that MRI has a greater than 97% negative predictive value for differentiating benign from malignant RSLs, suggesting that the absence of enhancement excludes malignancy (37,38).

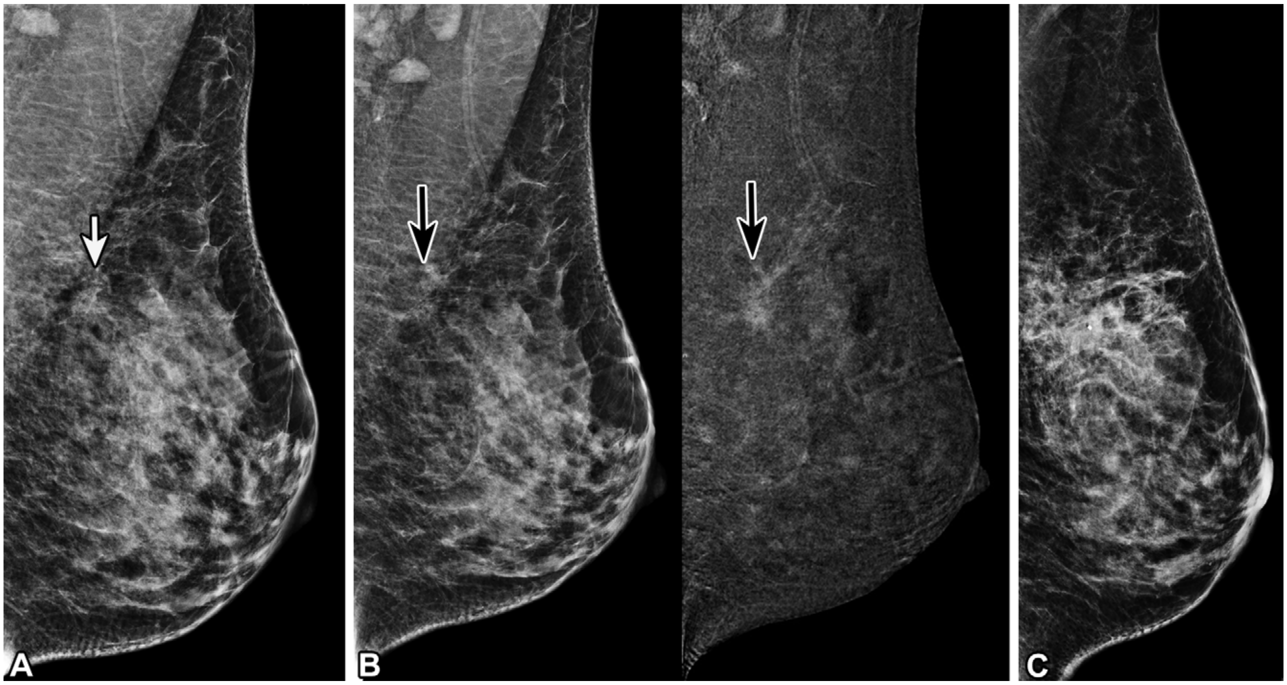


Figure 6. CSL in a 41-year-old woman. **(A)** Screening mediolateral oblique DBT mammogram of the left breast shows architectural distortion (arrow). The patient was recalled for contrast-enhanced mammography. **(B)** Mediolateral oblique contrast-enhanced mammogram (left) and recombined image (right) of the left breast show nonmass enhancement (arrow, left) **(C)** Mediolateral digital mammogram shows the biopsy clip at the site of architectural distortion. DBT-guided core biopsy was performed, revealing CSL, which was confirmed at subsequent surgical excision.

On the other hand, one study (39) of 175 women with mammographically detected architectural distortion showed a positive predictive value of only 30%, indicating that the presence of enhancement at MRI does not allow RSL to be distinguished from cancer (39). Other studies (37–40) have similarly demonstrated that MRI cannot reliably allow differentiation of RSLs that are associated with atypia or malignancy.

Contrast-enhanced Mammography.—To our knowledge, only two studies (41,42) have been performed to evaluate architectural distortion at contrast-enhanced mammography and its association with radial scars. Patel et al (41) demonstrated that 40% of radial scars were associated with enhancing distortion, whereas the remaining radial scars were nonenhancing. This study showed no identifying features that were used to differentiate benign from malignant enhancement and provided no data showing whether features at contrast-enhanced mammography could be used to predict upgrade rates. In a more recent trial (42), the authors evaluated architectural distortion at contrast-enhanced mammography but did not provide specific data on RSL (Fig 6). Additional data are needed as the use of this technology in clinical practice increases.

Management

Upgrade Rates

Reported upgrade rates of RSLs at CNB vary. In a large meta-analysis by Farshid and Buckley (43), 49 examinations of 3163 radial scars that were surgically excised were analyzed by sampling method and pathologic type with or without atypia (Table). The overall rate of upgrade to ductal carcinoma in situ (DCIS) or invasive carcinoma was 6.9% (Fig 7). Of those, 32.7% were upgraded to invasive carcinoma and 66.4% to DCIS (44). Several studies (44–46) have shown that carcinoma may be located peripheral and eccentric to the RSL (45–46). Histopathologic features associated with upgrading of RSLs include larger size of the lesion, the presence of atypical hyperplasia, and the presence of other coexisting high-risk lesions (14). In one study (25) of 146 RSLs shown at core biopsy among patients who had undergone DBT, 13.8% (four of 29) of RSLs with atypia were upgraded to malignancy at excision versus 0.9% (one of 117) of RSL without atypia. Of the upgraded RSLs with atypia, two had associated calcifications. Authors of another study found that RSLs with mass or distortion at digital mammography or US were more likely to be upgraded to malignancy than were RSLs manifesting as calcifications; however, distortion is less likely to be detected at digital mammography than at DBT, so these findings may not apply to DBT-detected distortion (47,48). Because of the increased detection of RSL with DBT and the same low upgrade rate of RSL to malignancy before and after the introduction of DBT (17), radial scars without atypia at biopsy of architectural distortion alone, without associated calcifications or a mass, may be appropriate for surveillance, particularly if the area of distortion is small. However, there is no consensus regarding surveillance management protocols when surveillance is chosen, and no evidence, to our knowledge, suggests the most appropriate follow-up interval.

Multiple studies of RSLs without atypia have shown upgrade rates ranging from 0% to 2.2% (25,49,50). The sampling method, number of samples, gauge of the needle, volume of tissue obtained, target being sampled (mass vs calcification vs architectural distortion), and quality of radiologic–pathologic

Summary of Studies of RSLs and Reported Upgrades to Malignancy Reported in a Meta-Analysis by Farshid and Buckley

	Core Needle Gauge		
Abnormality	14-Gauge	8–16-Gauge*	8–11-Gauge†
No. of RSLs without atypia	828	1263	122
No. upgraded to invasive cancer	18 (2.2)	10 (0.8)	0
No. upgraded to DCIS	30 (3.6)	29 (2.3)	2 (1.6)
Total no. of upgrades	48 (5.8)	39 (3.0)	2 (1.6)
RSLs, presence of atypia not specified or mixed	201	243	115
No. upgraded to invasive cancer	16 (8.0)	10 (4.1)	2 (1.7)
No. upgraded to DCIS	14 (7.0)	19 (7.8)	4 (3.5)
Total no. of upgrades	30 (14.9)	29 (12.0)	6 (5.2)
RSLs with atypia	114	171	11
No. upgraded to invasive cancer	5 (4.4)	7 (4.1)	NA
No. upgraded to DCIS	22 (19.3)	18 (10.5)	NA
Total no. of upgrades	27 (23.7)	25 (14.6)	2 (18.2)

Source.—Reference 43.
Note.— Data in parentheses are percentages. DCIS = ductal carcinoma in situ, NA = not available.
* Both vacuum-assisted and non–vacuum-assisted core biopsy.
† Vacuum-assisted core biopsy.

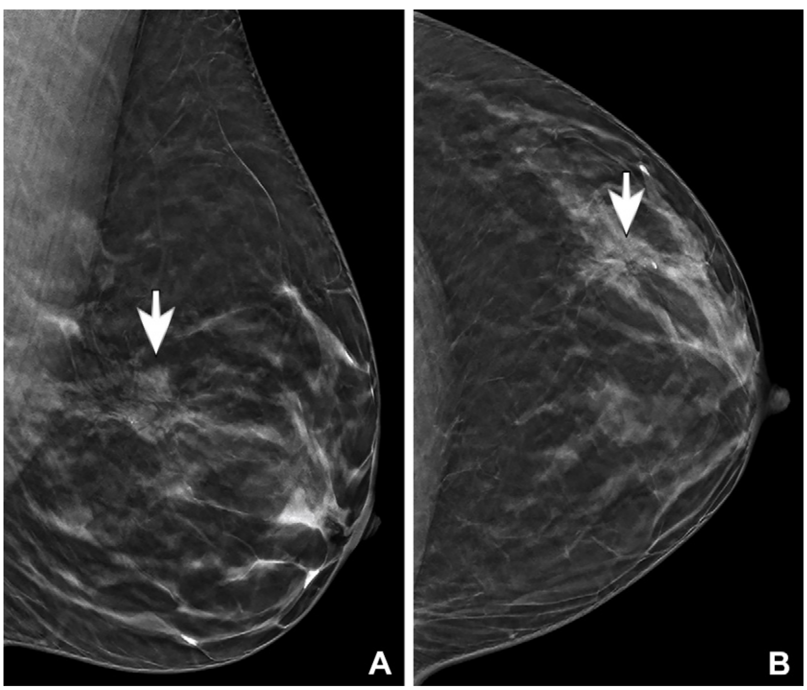


Figure 7. Radial scar with focal atypia and subsequent upgrade in a 48-year-old woman. Screening DBT mammograms show architectural distortion, which was first identified in the upper outer left breast (arrow). No correlate was found at targeted US (not shown). DBT-guided core biopsy showed a radial scar with focal atypia. Surgical excision revealed a 3-mm DCIS. The patient underwent lumpectomy and was treated with tamoxifen.

review (40) can affect the rate of upgrade to carcinoma. The vacuum-assisted biopsy sampling technique yields a greater volume of tissue relative to the volume of the overall lesion. As a result, radial scars without atypia that are biopsied with vacuum assistance had a rate of upgrade to DCIS of 1%. For radial scars biopsied with a 14-gauge core needle without vacuum assistance, the upgrade rates for those without and with associated atypia were 5% and 29%, respectively (43). In a retrospective study, Resetkova et al (51) assessed outcomes of patients who underwent workup using a 9–11-gauge stereotactic core needle, with a mean of 10 core specimens obtained. The authors accurately diagnosed benign versus atypical RSL in 17 of 19 cases (89%). In addition, no patients were upgraded from RSL to malignancy after surgical excision, and only two patients were upgraded from proliferative RSL to RSL with atypia. Although these study results suggest that a large number of core samples results in higher accuracy in diagnosis, there are risks of associated bleeding and infection, which may increase with each additional core sample obtained. A proposed management schematic is shown in Figure 8.

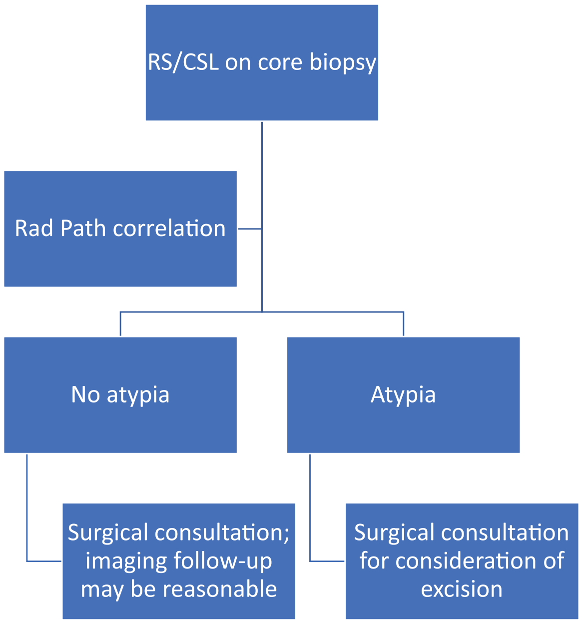


Figure 8. Proposed management algorithm for RSL. *Rad Path* = radiology–pathology.

Use of techniques that decrease upgrade rates may allow patients to undergo imaging follow-up to monitor change over time rather than undergoing invasive surgical excision.

Surgical Excision versus Surveillance

Management of RSL remains controversial, with no clear consensus. Authors of a previous review (22) suggested management based on the size of the imaging and pathologic findings. Although RSL without atypia is associated with low upgrade rates, women with RSL have a two-fold risk of breast carcinoma compared with women without RSL (52). This may be attributable to the fact that RSL often coexists with other high-risk proliferative lesions that have intrinsic malignant potential independent of RSL. As such, surgical excision of RSL was previously routinely performed to exclude malignancy, regardless of the presence of atypia. However, the practice is evolving with a multidisciplinary approach to the management of RSL. Individual patient risk factors and histopathologic and imaging findings are considered, and, at some centers, lack of enhancement at MRI may support a decision to undergo imaging surveillance rather than surgical excision, although this approach is based on data from few and small studies. The American Society of Breast Surgeons (53) continues to recommend excision for the larger CSL but allows imaging follow-up for smaller radial scars without atypia that were thought to be well sampled at biopsy with radiologic–pathologic concordance. Finally, although women who have RSLs, with or without atypia, may have a slightly increased risk for breast cancer, they are not considered to be at high enough risk to recommend increased surveillance or supplemental screening.

Conclusion

RSLs are more commonly encountered as increased use of screening DBT allows identification of more architectural distortion. Studies have shown that accurate imaging-guided biopsy with larger tissue samples yields lower upgrade rates and that RSLs without associated atypia have low (0%–2.2%) upgrade rates at surgery. Careful radiologic–pathologic review and consideration of patient risk factors are critical to optimal management recommendations. Some centers use a multidisciplinary approach to the management of RSLs, allowing a patient-centered discussion regarding surgical excision versus imaging surveillance. Individual patient risk factors and histopathologic and imaging findings are considered, and, in some centers, lack of enhancement at MRI may support a decision to undergo imaging surveillance rather than surgical excision. Breast radiologists play a critical role in patient treatment recommendations and must be mindful that surgical excision for all RSLs, particularly those without associated atypia, may result in overtreatment of patients. Thus, imaging follow-up may be a reasonable approach for patients with accurately sampled RSLs without atypia.

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